BUSINESS EXCELLENCE ANALYTICS REPORT

(Furniture, Office Supplies & Technology Performance Analysis)

1. Project Overview and Objective

The dataset contains common data quality issues such as missing values, duplicate records, incorrect data entries, and inconsistent formatting. Using Excel, Power Query, and Power BI, the data is cleaned, transformed, and analyzed. Since the dataset consists of multiple related tables, data modeling is performed by creating relationships between the tables. other visualizations are used to build an interactive dashboard for sales analysis and business decision-making.

2. Data Sources

- **Source Description and Timeline:** Google Dataset Search and (2023 - 2025)
- **Domain**: Retail

3. Problem Statement

This project analyzing Retail Sales data from 2023 to 2025 to identify sales trends, customer purchasing patterns, product performance, and profit insights

4. Attribute (Column /Features) Details:

Sales Sheet 1:

Attribute Name	Data Type	Description
Order ID	Integer / String	Unique identifier for each orders
Order Date	Date Data Type	Order Date for each orders
Ship Date	Date Data Type	Shipped date for each orders
Customer ID	Integer / String	Unique identifier for each Customers
Product ID	Integer / String	Unique identifier for each product
Sales	Decimal Number	Sales amount for each sales
Quantity	Whole Number	Quantity for each Orders
Discount	Decimal Number	Discount for each Orders
Ship Mode	Text/Categorical	Shipping mode used for deliver the order
Sales Channel	Text	Sales channel used for order
Returned	Categorical	Shows whether the sale was returned

Customer Sheet 2:

Attribute Name	Data Type	Description
Customer ID	Integer / String	Unique identifier for each Customers
Segment	Categorical	Segment for each customer type
State	Categorical	State of the each customer
Customer Rating	Categorical	Rating given by the customer

Product Sheet 3:

Attribute Name	Data Type	Description
Product ID	Integer / String	Unique identifier for each product
Category	Categorical	Category for each product
Sub-Category	Categorical	Sub Category for each product
Product Name	Text	Name of each Product

5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

1.Data Cleaning & Transformation: Raw data changed to standardized and structured table format

Order ID	Order Dat	Ship Dat	Customer I	Product I	Sales	Quantit	Discoun	Ship Mode	Sales Channe	Returne
ORD00001	01/26/2023	2023-01-29	CUST0211	PROD002	₹ 4,075	6	0.2		mobile App	yes
ORD00002	10/25/2023	2023-10-29	CUST0178	PROD049	₹ 4,035	5	0	First Class	mobile App	no
ORD00003	25-11-2024	2024-11-30	CUST0069	PROD053	₹ 1,676	10	0.1		Mobile App	yes
ORD00004	12/01/2024	2024-12-04	CUST0053	PROD080	₹ 4,538	3	0.3	First Class	Online	yes
ORD00005	11/16/2023	2023-11-18	CUST0157	PROD004	₹ 783	7	0.1	second Class	store	no
ORD00006	25-11-2024	2024-11-30	CUST0118	PROD012	₹ 1,026	9	0.1	Standard Class	Store	no
ORD00007	01/07/2025	2025-01-10	CUST0202	PROD007	₹ 352	2	0	second Class	Store	yes
ORD00008	09/20/2024	2024-09-21	CUST0186	PROD036	₹ 2,064	3	0.3	second Class	Store	no
ORD00009	02-08-2023	2023-08-09	CUST0020	PROD022	₹ 3,637	5	0.2	Standard Class	Store	no
ORD00010	07/16/2023	2023-07-17	CUST0177	PROD028	₹ 569	8	0.1	second Class	online	no
ORD00011	01/11/2023	2023-01-16	CUST0004	PROD068	₹ 4,248	4	0.3		Mobile App	no
ORD00012	26-03-2024	2024-03-30	CUST0130	PROD077	₹ 1,533	1	0.1	First Class	Store	no
ORD00013	03/06/2025	2025-03-12	CUST0027	PROD085	₹ 1,616	5	0.1	second Class	Store	yes
ORD00014	02/27/2024	2024-02-29	CUST0028	PROD035	₹ 2,865	3	0.3	First Class	Online	yes
ORD00015	05-05-2024	2024-05-08	CUST0223	PROD038	₹ 3,781	10	0.1	second Class	online	no
ORD00016	09/15/2023	2023-09-21	CUST0223	PROD009	₹ 827	6	0.2	second Class	Mobile App	no
ORD00017	10/05/2023	2023-10-06	CUST0239	PROD066	₹ 4,480	10	0	second Class	online	no
ORD00018	15-04-2025	2025-04-20	CUST0006	PROD042	₹ 543	3	0.1	First Class	Mobile App	no
ORD00019	04/23/2023	2023-04-30	CUST0135	PROD075	₹ 2,608	2	0.1	second Class	Mobile App	yes

Sales | Customers | Products

Customer II	Segment	State	Customer Ratin
CUST0001	Home Office	MAHARASHTRA	4
CUST0002	consumer	telangana	1
CUST0003	Consumer	tElanganA	5
CUST0004	Consumer	Kerala	3
CUST0005	home Office	maharashtra	4
CUST0006	Corporate	Tamil Nadu	4
CUST0007	Consumer	TAMIL NADU	3
CUST0008	Home Office	karnataka	2
CUST0009	Home Office	Tamil Nadu	5
CUST0010	Corporate	Tamil Nadu	5
CUST0011	consumer	Karnataka	4
CUST0012	Home Office	Kerala	1
CUST0013	Home Office	Maharashtra	2
CUST0014	corporate	Karnataka	1
CUST0015	Corporate	Maharashtra	2
CUST0016	Consumer	Karnataka	3
CUST0017	corporate	Telangana	5
CUST0018	Consumer	Maharashtra	3
CUST0019	Home Office	Tamil Nadu	3

Sales | Customers | Products

Product I	Category	Sub-Categor	Product Nam
PROD001	Office Supplies	Accessories	Product 1
PROD002	Office Supplies	Binders	Product 2
PROD003	Furniture	Chairs	Product 3
PROD004	Technology	Binders	Product 4
PROD005	Technology	Storage	Product 5
PROD006	Furniture	Binders	Product 6
PROD007	Furniture	Storage	Product 7
PROD008	Furniture	Phones	Product 8
PROD009	Technology	Storage	Product 9
PROD010	Furniture	Tables	Product 10
PROD011	Office Supplies	Chairs	Product 11
PROD012	Office Supplies	Chairs	Product 12
PROD013	Furniture	Accessories	Product 13
PROD014	Technology	Chairs	Product 14
PROD015	Furniture	Tables	Product 15
PROD016	Office Supplies	Storage	Product 16
PROD017	Furniture	Binders	Product 17
PROD018	Office Supplies	Accessories	Product 18
PROD019	Office Supplies	Accessories	Product 19

Sales Customers Products +

2.Data Cleaning:

From sales table I have used **LEFT,RIGHT,MID** functions on the **Order ID** Column to Extract the Id text into 3 separate columns

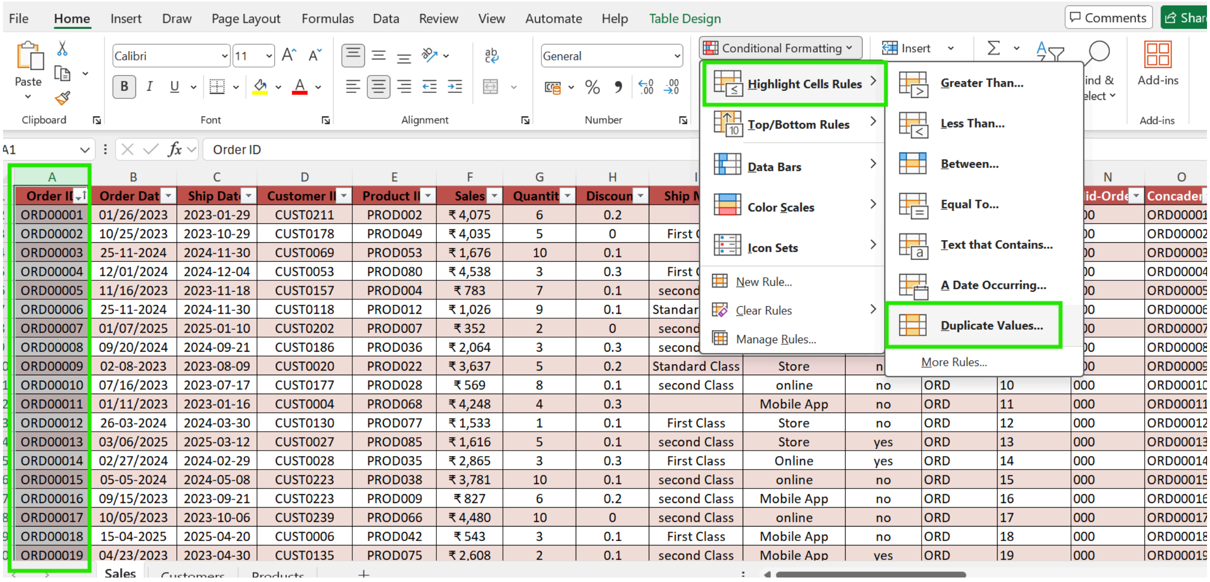
Next I have applied **CONCATENATE** Functions on the those **extracted separated** columns merged to one single column

Then used the **PROPER** functions on the **RETURNED, SALES CHANNEL,SHIP MODE** Columns to correct the text formatting and capitalize the first letter of each word

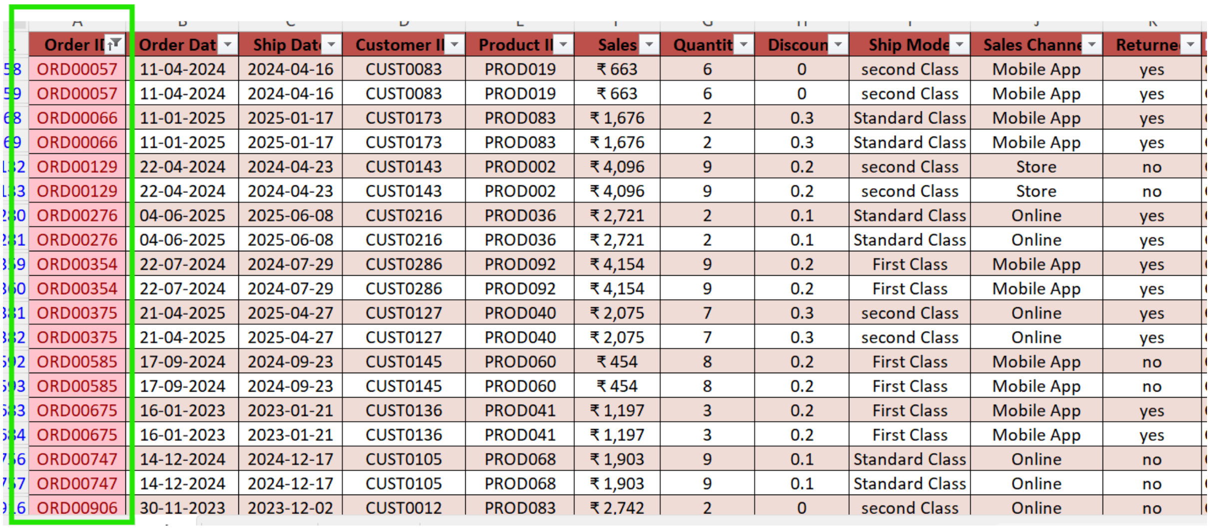
L	M	N	O	P	Q	R	
Left -Orde	Right - Or	Mid-Orde	Concade	Proper -Returned	Proper - Sales Channel	Proper-Ship mode	C
ORD	01	000	ORD00001	Yes	Mobile App		
ORD	02	000	ORD00002	No	Mobile App	First Class	
ORD	03	000	ORD00003	Yes	Mobile App		
ORD	04	000	ORD00004	Yes	Online	First Class	
ORD	05	000	ORD00005	No	Store	Second Class	
ORD	06	000	ORD00006	No	Store	Standard Class	
ORD	07	000	ORD00007	Yes	Store	Second Class	
ORD	08	000	ORD00008	No	Store	Second Class	
ORD	09	000	ORD00009	No	Store	Standard Class	
ORD	10	000	ORD00010	No	Online	Second Class	
ORD	11	000	ORD00011	No	Mobile App		
ORD	12	000	ORD00012	No	Store	First Class	
ORD	13	000	ORD00013	Yes	Store	Second Class	
ORD	14	000	ORD00014	Yes	Online	First Class	
ORD	15	000	ORD00015	No	Online	Second Class	
ORD	16	000	ORD00016	No	Mobile App	Second Class	
ORD	17	000	ORD00017	No	Online	Second Class	
ORD	18	000	ORD00018	No	Mobile App	First Class	
ORD	19	000	ORD00019	Yes	Mobile App	Second Class	

Find Duplicate values

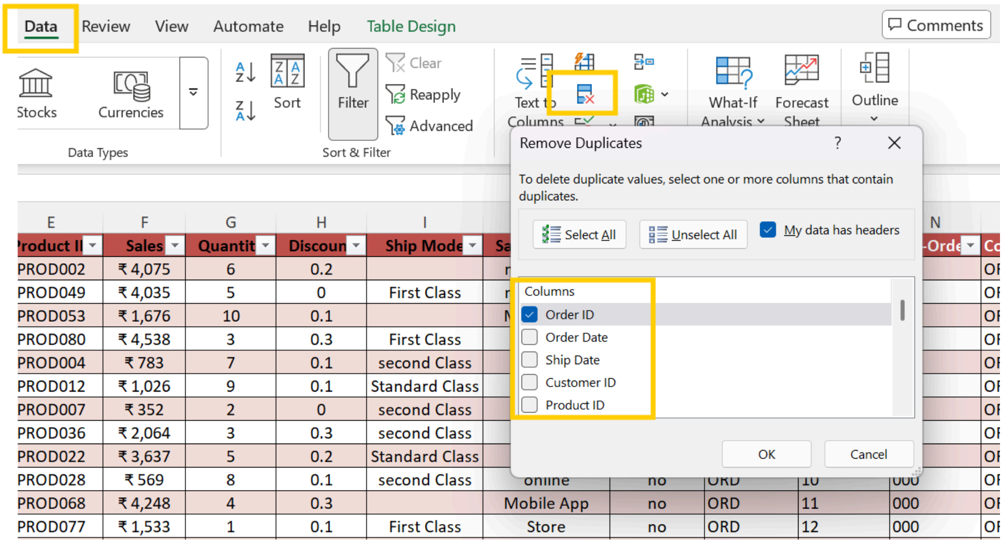
In the Sales table, I used **Conditional Formatting** and applied the **Duplicate Values** option to the Order ID column to identify duplicate orders

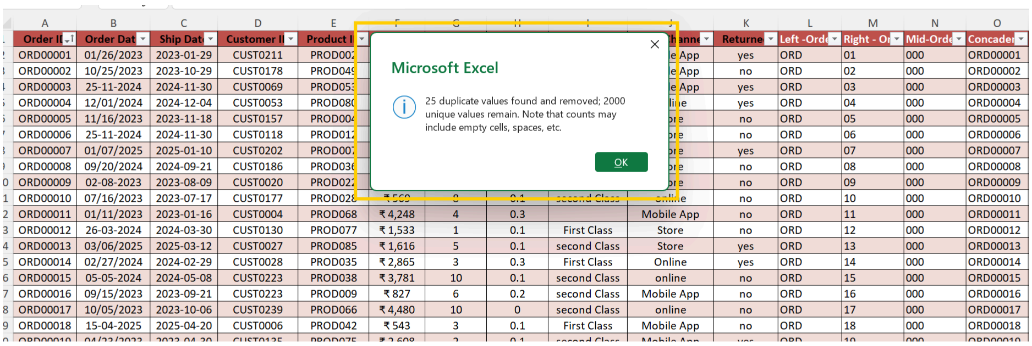


Then Filtered the duplicate **Order ID**s using the **Filter by Color option**



Order II	Order Dat	Ship Dat	Customer II	Product II	Sales	Quantit	Discoun	Ship Mode	Sales Channe	Returne
ORD00057	11-04-2024	2024-04-16	CUST0083	PROD019	₹ 663	6	0	second Class	Mobile App	yes
ORD00057	11-04-2024	2024-04-16	CUST0083	PROD019	₹ 663	6	0	second Class	Mobile App	yes
ORD00066	11-01-2025	2025-01-17	CUST0173	PROD083	₹ 1,676	2	0.3	Standard Class	Mobile App	yes
ORD00066	11-01-2025	2025-01-17	CUST0173	PROD083	₹ 1,676	2	0.3	Standard Class	Mobile App	yes
ORD00129	22-04-2024	2024-04-23	CUST0143	PROD002	₹ 4,096	9	0.2	second Class	Store	no
ORD00129	22-04-2024	2024-04-23	CUST0143	PROD002	₹ 4,096	9	0.2	second Class	Store	no
ORD00276	04-06-2025	2025-06-08	CUST0216	PROD036	₹ 2,721	2	0.1	Standard Class	Online	yes
ORD00276	04-06-2025	2025-06-08	CUST0216	PROD036	₹ 2,721	2	0.1	Standard Class	Online	yes
ORD00354	22-07-2024	2024-07-29	CUST0286	PROD092	₹ 4,154	9	0.2	First Class	Mobile App	yes
ORD00354	22-07-2024	2024-07-29	CUST0286	PROD092	₹ 4,154	9	0.2	First Class	Mobile App	yes
ORD00375	21-04-2025	2025-04-27	CUST0127	PROD040	₹ 2,075	7	0.3	second Class	Online	yes
ORD00375	21-04-2025	2025-04-27	CUST0127	PROD040	₹ 2,075	7	0.3	second Class	Online	yes
ORD00585	17-09-2024	2024-09-23	CUST0145	PROD060	₹ 454	8	0.2	First Class	Mobile App	no
ORD00585	17-09-2024	2024-09-23	CUST0145	PROD060	₹ 454	8	0.2	First Class	Mobile App	no
ORD00675	16-01-2023	2023-01-21	CUST0136	PROD041	₹ 1,197	3	0.2	First Class	Mobile App	yes
ORD00675	16-01-2023	2023-01-21	CUST0136	PROD041	₹ 1,197	3	0.2	First Class	Mobile App	yes
ORD00747	14-12-2024	2024-12-17	CUST0105	PROD068	₹ 1,903	9	0.1	Standard Class	Online	no
ORD00747	14-12-2024	2024-12-17	CUST0105	PROD068	₹ 1,903	9	0.1	Standard Class	Online	no
ORD00906	30-11-2023	2023-12-02	CUST0012	PROD083	₹ 2,742	2	0	second Class	Online	no





In the **Data ribbon,** I applied the **Duplicate Values** option to the Order ID column. I identified **25 duplicate values** and removed them.

3.Formatting Date Properly

In the Sales table, the **Order Date** and **Ship Date** were not formatted properly. I applied the Delimiter option and used the **DATE, IF,** and **VALUE** functions to convert the dates into the proper **DD-MM-YYYY** format. Now, both the Order Date and Ship Date are cleaned properly

Order Date formatted	Ship date formatted	Proper Order Date	Proper ship date
01-26-2023	2023-01-29	26-01-2023	29-01-2023
10-25-2023	2023-10-29	25-10-2023	29-10-2023
25-11-2024	2024-11-30	25-11-2024	30-11-2024
12-01-2024	2024-12-04	01-12-2024	04-12-2024
11-16-2023	2023-11-18	16-11-2023	18-11-2023
25-11-2024	2024-11-30	25-11-2024	30-11-2024
01-07-2025	2025-01-10	07-01-2025	10-01-2025
09-20-2024	2024-09-21	20-09-2024	21-09-2024
02-08-2023	2023-08-09	08-02-2023	09-08-2023
07-16-2023	2023-07-17	16-07-2023	17-07-2023
01-11-2023	2023-01-16	11-01-2023	16-01-2023
26-03-2024	2024-03-30	26-03-2024	30-03-2024
03-06-2025	2025-03-12	06-03-2025	12-03-2025
02-27-2024	2024-02-29	27-02-2024	29-02-2024
05-05-2024	2024-05-08	05-05-2024	08-05-2024
09-15-2023	2023-09-21	15-09-2023	21-09-2023
10-05-2023	2023-10-06	05-10-2023	06-10-2023
15-04-2025	2025-04-20	15-04-2025	20-04-2025

Clipboard Font Alignment Number Styles Cells Editing Add-ins

U2

```
=DATE(RIGHT([@[Order Date formatted]],4),IF(VALUE(LEFT([@[Order Date formatted]],2))>12,VALUE(MID([@[Order Date formatted]],4,2)),VALUE(LEFT([@[Order Date formatted]],2))),IF(VALUE(LEFT([@[Order Date formatted]],2))>12,VALUE(LEFT([@[Order Date formatted]],2)),VALUE(MID([@[Order Date formatted]],4,2))))
```

	Q	R	S	T	U	V	W	X	Y	Z
1	Proper - Sales Channel	Proper-Ship mode	Order Date formatted	Ship date formatted	Proper Order Date	Proper ship date	Column13	Column14	Column15	Column16
2	Mobile App		01-26-2023	2023-01-29	26-01-2023	29-01-2023				
3	Mobile App	First Class	10-25-2023	2023-10-29	25-10-2023	29-10-2023				
4	Mobile App		25-11-2024	2024-11-30	25-11-2024	30-11-2024				
5	Online	First Class	12-01-2024	2024-12-04	01-12-2024	04-12-2024				
6	Store	Second Class	11-16-2023	2023-11-18	16-11-2023	18-11-2023				
7	Store	Standard Class	25-11-2024	2024-11-30	25-11-2024	30-11-2024				
8	Store	Second Class	01-07-2025	2025-01-10	07-01-2025	10-01-2025				
9	Store	Second Class	09-20-2024	2024-09-21	20-09-2024	21-09-2024				
10	Store	Standard Class	02-08-2023	2023-08-09	08-02-2023	09-08-2023				
11	Online	Second Class	07-16-2023	2023-07-17	16-07-2023	17-07-2023				
12	Mobile App		01-11-2023	2023-01-16	11-01-2023	16-01-2023				
13	Store	First Class	26-03-2024	2024-03-30	26-03-2024	30-03-2024				
14	Store	Second Class	03-06-2025	2025-03-12	06-03-2025	12-03-2025				
15	Online	First Class	02-27-2024	2024-02-29	27-02-2024	29-02-2024				
16	Online	Second Class	05-05-2024	2024-05-08	05-05-2024	08-05-2024				
17	Mobile App	Second Class	09-15-2023	2023-09-21	15-09-2023	21-09-2023				

4.Added Columns

Then, in the Customer table, I applied Nested IF conditions to the Customer Rating column. I categorized the ratings as follows: 5 – Excellent, 4 – Good, 3 – Average, 2 – Poor, and 1 – Very Poor

E8

```
=IF([@[Customer Rating]]>4,"Excellent",IF([@[Customer Rating]]>3,"Good",IF([@[Customer Rating]]>2, "Average",IF([@[Customer Rating]]>1,"Poor","Very Poor"))))
```

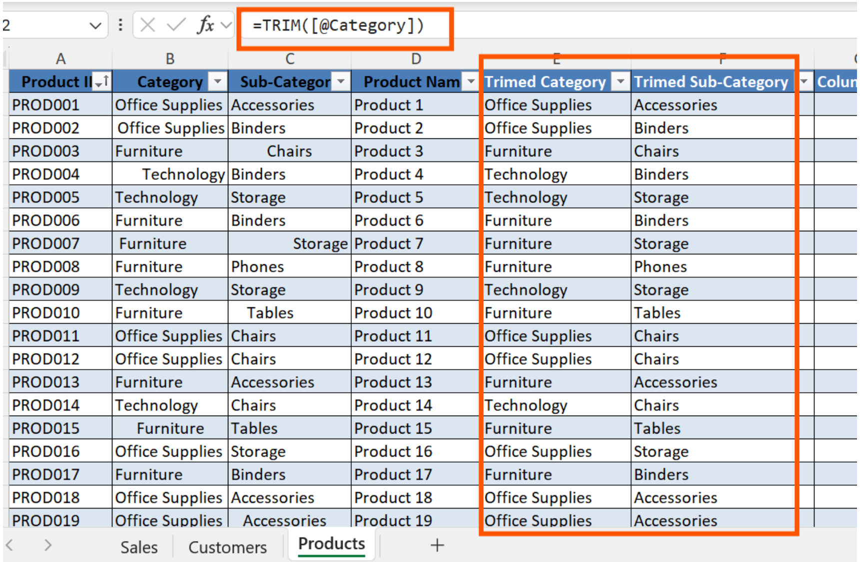
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Customer II	Segment	State	Customer Rating	Rating status	Proper Segement	Proper State	Column4	Column5	Column6	Column7	Column8	Column9
2	CUST0001	Home Office	MAHARASHTRA	4	Good	Home Office	Maharashtra						
3	CUST0002	consumer	telangana	1	Very Poor	Consumer	Telangana						
4	CUST0003	Consumer	tElanganA	5	Excellent	Consumer	Telangana						
5	CUST0004	Consumer	Kerala	3	Average	Consumer	Kerala						
6	CUST0005	home Office	maharashtra	4	Good	Home Office	Maharashtra						
7	CUST0006	Corporate	Tamil Nadu	4	Good	Corporate	Tamil Nadu						
8	CUST0007	Consumer	TAMIL NADU	3	Average	Consumer	Tamil Nadu						
9	CUST0008	Home Office	karnataka	2	Poor	Home Office	Karnataka						
10	CUST0009	Home Office	Tamil Nadu	5	Excellent	Home Office	Tamil Nadu						
11	CUST0010	Corporate	Tamil Nadu	5	Excellent	Corporate	Tamil Nadu						
12	CUST0011	consumer	Karnataka	4	Good	Consumer	Karnataka						
13	CUST0012	Home Office	Kerala	1	Very Poor	Home Office	Kerala						
14	CUST0013	Home Office	Maharashtra	2	Poor	Home Office	Maharashtra						
15	CUST0014	corporate	Karnataka	1	Very Poor	Corporate	Karnataka						
16	CUST0015	Corporate	Maharashtra	2	Poor	Corporate	Maharashtra						
17	CUST0016	Consumer	Karnataka	3	Average	Consumer	Karnataka						
18	CUST0017	corporate	Telangana	5	Excellent	Corporate	Telangana						

Sales Customers Products

Then, the text in the **Segment and State columns** was not formatted properly, so I used the **PROPER** function to correct the text formatting.

5.Extra Spaces and Inconsistencies

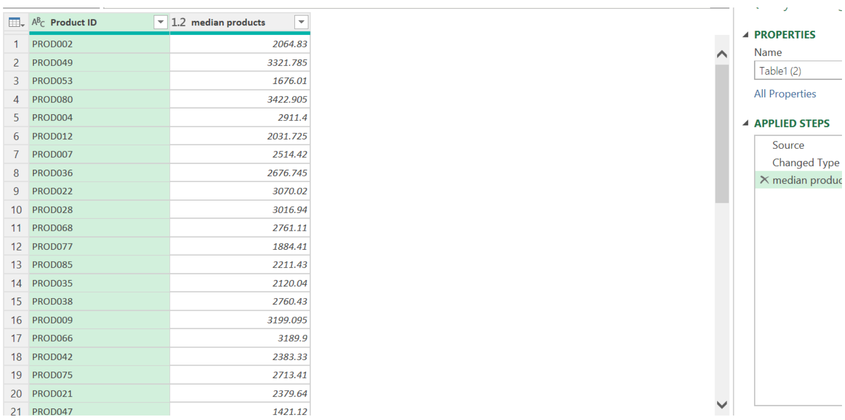
In the Products table, the **Category and Sub-Category columns** text had extra spaces and inconsistencies. So, I used the **TRIM** function to remove the extra spaces and correct the text.



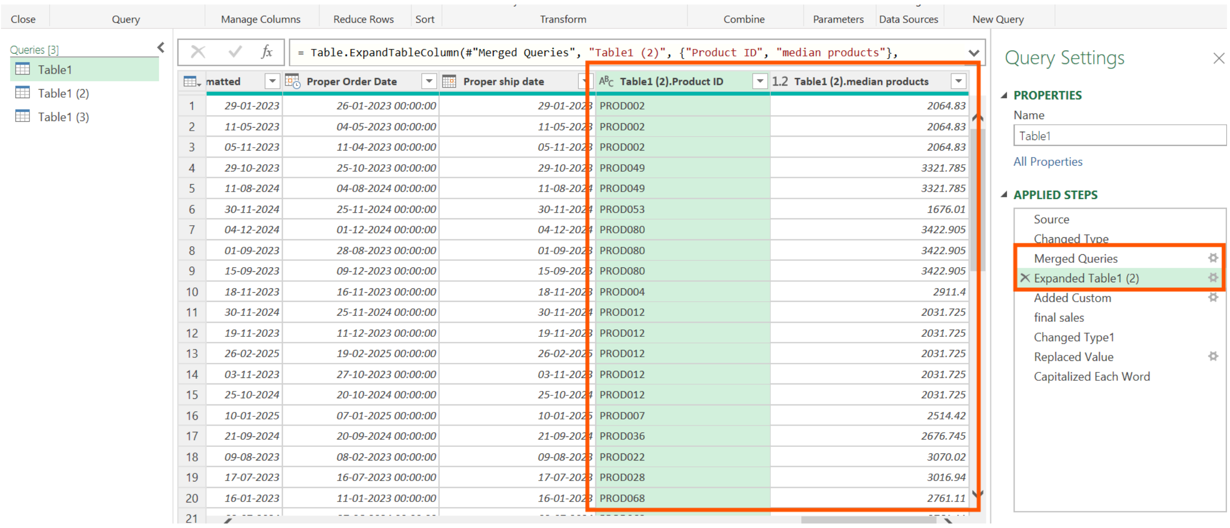
Product ID	Category	Sub-Category	Product Name	Trimed Category	Trimed Sub-Category
PROD001	Office Supplies	Accessories	Product 1	Office Supplies	Accessories
PROD002	Office Supplies	Binders	Product 2	Office Supplies	Binders
PROD003	Furniture	Chairs	Product 3	Furniture	Chairs
PROD004	Technology	Binders	Product 4	Technology	Binders
PROD005	Technology	Storage	Product 5	Technology	Storage
PROD006	Furniture	Binders	Product 6	Furniture	Binders
PROD007	Furniture	Storage	Product 7	Furniture	Storage
PROD008	Furniture	Phones	Product 8	Furniture	Phones
PROD009	Technology	Storage	Product 9	Technology	Storage
PROD010	Furniture	Tables	Product 10	Furniture	Tables
PROD011	Office Supplies	Chairs	Product 11	Office Supplies	Chairs
PROD012	Office Supplies	Chairs	Product 12	Office Supplies	Chairs
PROD013	Furniture	Accessories	Product 13	Furniture	Accessories
PROD014	Technology	Chairs	Product 14	Technology	Chairs
PROD015	Furniture	Tables	Product 15	Furniture	Tables
PROD016	Office Supplies	Storage	Product 16	Office Supplies	Storage
PROD017	Furniture	Binders	Product 17	Furniture	Binders
PROD018	Office Supplies	Accessories	Product 18	Office Supplies	Accessories
PROD019	Office Supplies	Accessories	Product 19	Office Supplies	Accessories

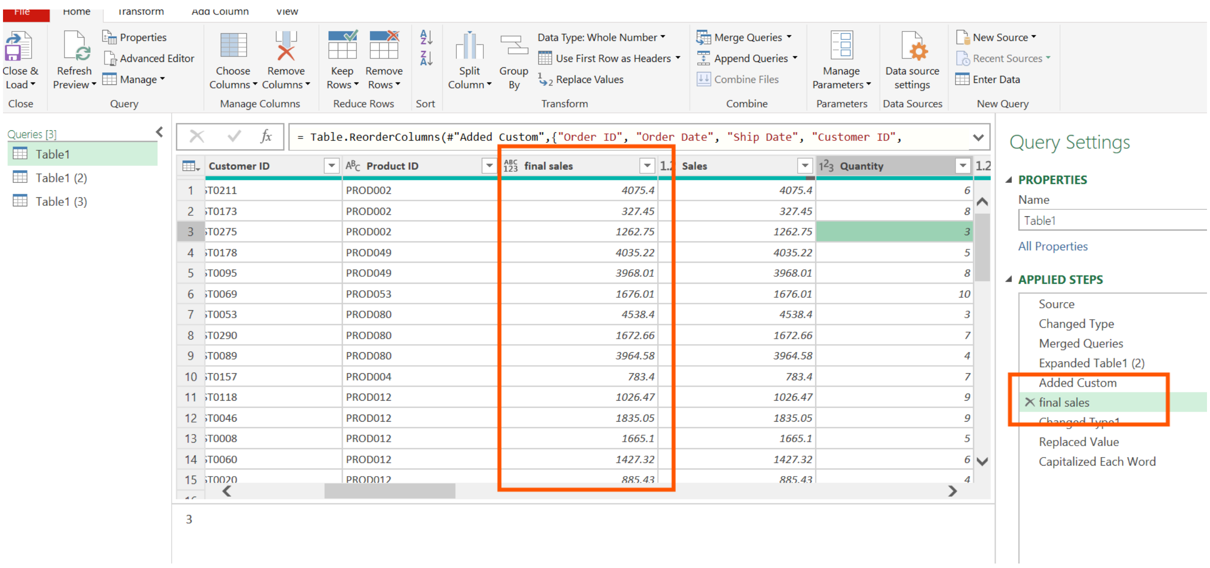
6.Power Querry

Created a product-wise median Sales table.Merged it with the Sales table using Product ID.Expanded the median Sales column.Used a Custom Column to fill the missing Sales values with the median.

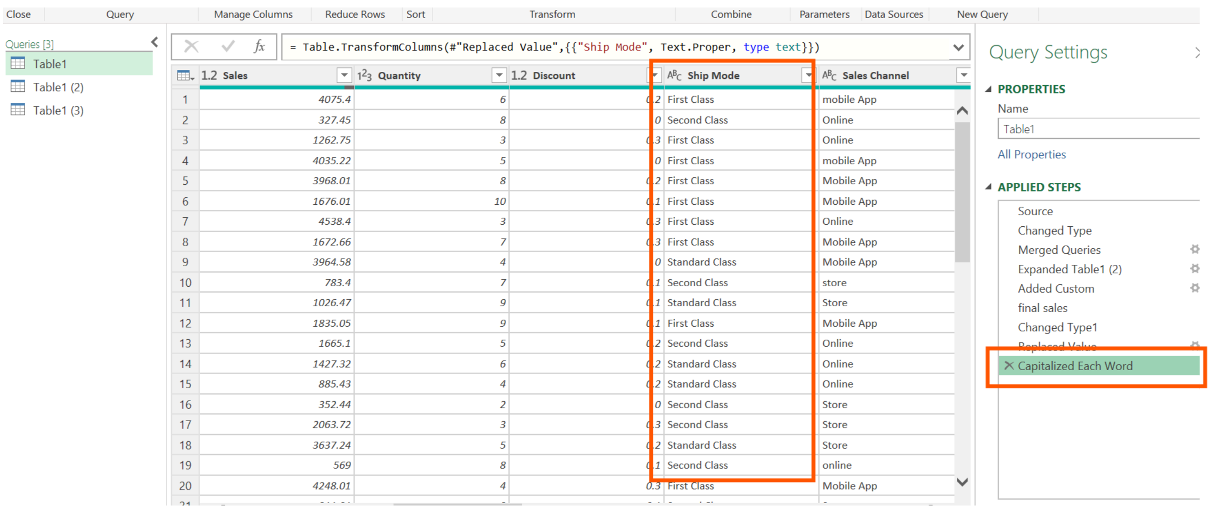


	Product ID	median products
1	PROD002	2064.83
2	PROD049	3321.785
3	PROD053	1676.01
4	PROD080	3422.905
5	PROD004	2911.4
6	PROD012	2031.725
7	PROD007	2514.42
8	PROD036	2676.745
9	PROD022	3070.02
10	PROD028	3016.94
11	PROD068	2761.11
12	PROD077	1884.41
13	PROD085	2211.43
14	PROD035	2120.04
15	PROD038	2760.43
16	PROD009	3199.095
17	PROD066	3189.9
18	PROD042	2383.33
19	PROD075	2713.41
20	PROD021	2379.64
21	PROD047	1421.12

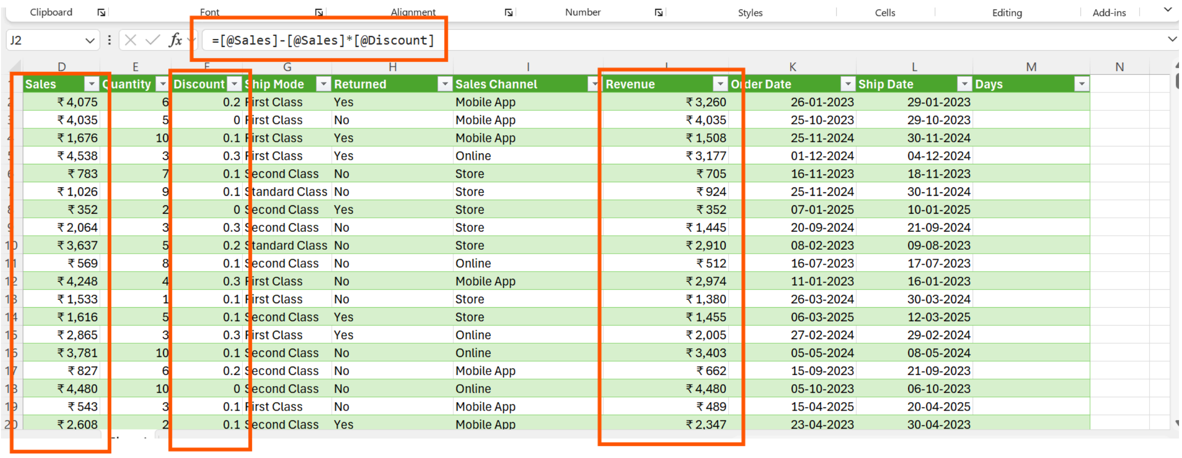




In the Ship Mode column, I identified the null values and replaced them using the mode value, which was First Class



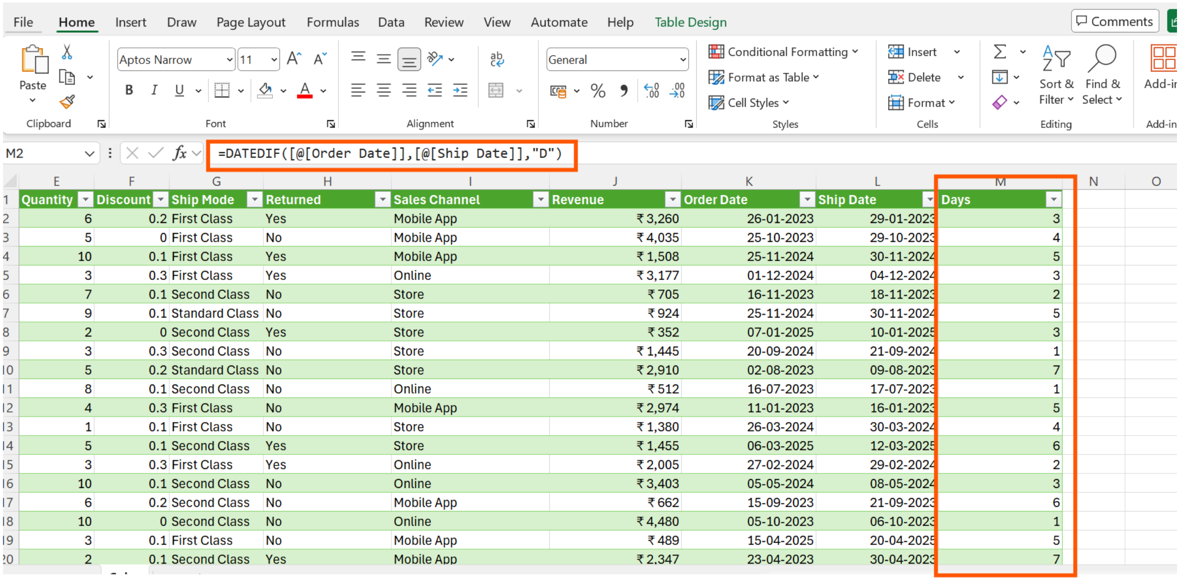
7.I calculated the **Revenue** using the **Sales Amount and Discount columns.**



J2 =[@Sales]-[@Sales]*[@Discount]

Sales	Quantity	Discount	Ship Mode	Returned	Sales Channel	Revenue	Order Date	Ship Date	Days
₹ 4,075	6	0.2	First Class	Yes	Mobile App	₹ 3,260	26-01-2023	29-01-2023	
₹ 4,035	5	0	First Class	No	Mobile App	₹ 4,035	25-10-2023	29-10-2023	
₹ 1,676	10	0.1	First Class	Yes	Mobile App	₹ 1,508	25-11-2024	30-11-2024	
₹ 4,538	3	0.3	First Class	Yes	Online	₹ 3,177	01-12-2024	04-12-2024	
₹ 783	7	0.1	Second Class	No	Store	₹ 705	16-11-2023	18-11-2023	
₹ 1,026	9	0.1	Standard Class	No	Store	₹ 924	25-11-2024	30-11-2024	
₹ 352	2	0	Second Class	Yes	Store	₹ 352	07-01-2025	10-01-2025	
₹ 2,064	3	0.3	Second Class	No	Store	₹ 1,445	20-09-2024	21-09-2024	
₹ 3,637	5	0.2	Standard Class	No	Store	₹ 2,910	08-02-2023	09-08-2023	
₹ 569	8	0.1	Second Class	No	Online	₹ 512	16-07-2023	17-07-2023	
₹ 4,248	4	0.3	First Class	No	Mobile App	₹ 2,974	11-01-2023	16-01-2023	
₹ 1,533	1	0.1	First Class	No	Store	₹ 1,380	26-03-2024	30-03-2024	
₹ 1,616	5	0.1	Second Class	Yes	Store	₹ 1,455	06-03-2025	12-03-2025	
₹ 2,865	3	0.3	First Class	Yes	Online	₹ 2,005	27-02-2024	29-02-2024	
₹ 3,781	10	0.1	Second Class	No	Online	₹ 3,403	05-05-2024	08-05-2024	
₹ 827	6	0.2	Second Class	No	Mobile App	₹ 662	15-09-2023	21-09-2023	
₹ 4,480	10	0	Second Class	No	Online	₹ 4,480	05-10-2023	06-10-2023	
₹ 543	3	0.1	First Class	No	Mobile App	₹ 489	15-04-2025	20-04-2025	
₹ 2,608	2	0.1	Second Class	Yes	Mobile App	₹ 2,347	23-04-2023	30-04-2023	

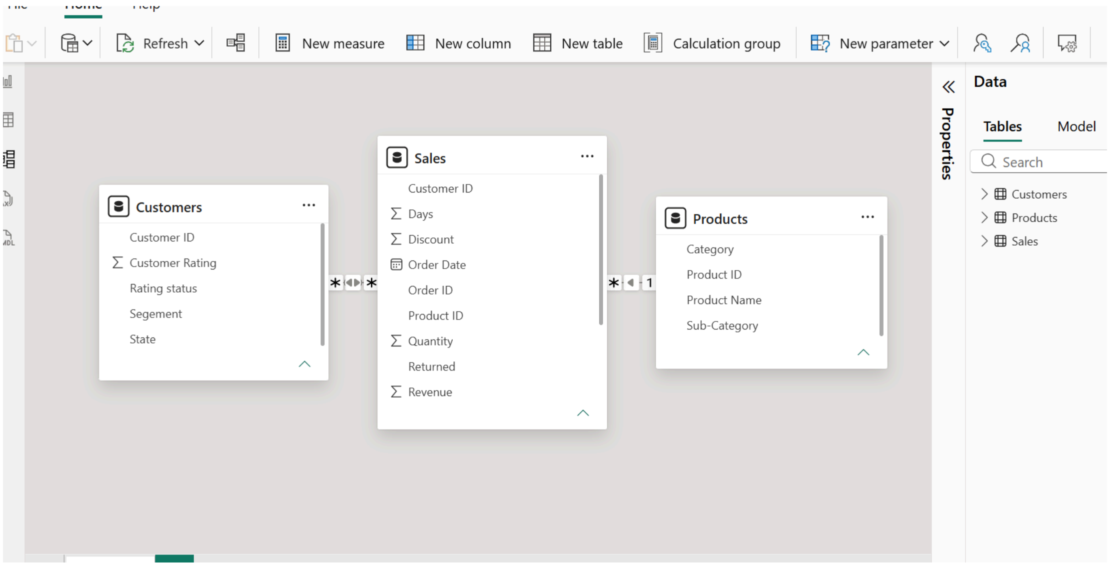
Finally I have calculated the number of days between the Order Date and Ship Date to determine the shipping duration



M2 =DATEDIF([@[Order Date]],[@[Ship Date]],"D")

Quantity	Discount	Ship Mode	Returned	Sales Channel	Revenue	Order Date	Ship Date	Days
6	0.2	First Class	Yes	Mobile App	₹ 3,260	26-01-2023	29-01-2023	3
5	0	First Class	No	Mobile App	₹ 4,035	25-10-2023	29-10-2023	4
10	0.1	First Class	Yes	Mobile App	₹ 1,508	25-11-2024	30-11-2024	5
3	0.3	First Class	Yes	Online	₹ 3,177	01-12-2024	04-12-2024	3
7	0.1	Second Class	No	Store	₹ 705	16-11-2023	18-11-2023	2
9	0.1	Standard Class	No	Store	₹ 924	25-11-2024	30-11-2024	5
2	0	Second Class	Yes	Store	₹ 352	07-01-2025	10-01-2025	3
3	0.3	Second Class	No	Store	₹ 1,445	20-09-2024	21-09-2024	1
5	0.2	Standard Class	No	Store	₹ 2,910	02-08-2023	09-08-2023	7
8	0.1	Second Class	No	Online	₹ 512	16-07-2023	17-07-2023	1
4	0.3	First Class	No	Mobile App	₹ 2,974	11-01-2023	16-01-2023	5
1	0.1	First Class	No	Store	₹ 1,380	26-03-2024	30-03-2024	4
5	0.1	Second Class	Yes	Store	₹ 1,455	06-03-2025	12-03-2025	6
3	0.3	First Class	Yes	Online	₹ 2,005	27-02-2024	29-02-2024	2
10	0.1	Second Class	No	Online	₹ 3,403	05-05-2024	08-05-2024	3
6	0.2	Second Class	No	Mobile App	₹ 662	15-09-2023	21-09-2023	6
10	0	Second Class	No	Online	₹ 4,480	05-10-2023	06-10-2023	1
3	0.1	First Class	No	Mobile App	₹ 489	15-04-2025	20-04-2025	5
2	0.1	Second Class	Yes	Mobile App	₹ 2,347	23-04-2023	30-04-2023	7

8 Power BI: Data modelling, DAX calculations, visualization, and interactive dashboard creation.



First, I created data relationships between the Customer, Sales, and Product tables in Power BI and checked that the relationships were active

I have created new Cards using measures to display Total Revenue, Average Shipping Days, Total orders and Total Quantity Sold

Created a pie chart to show the revenue distribution across different states

Created a clustered column chart to analyze revenue across different categories and sub-categories

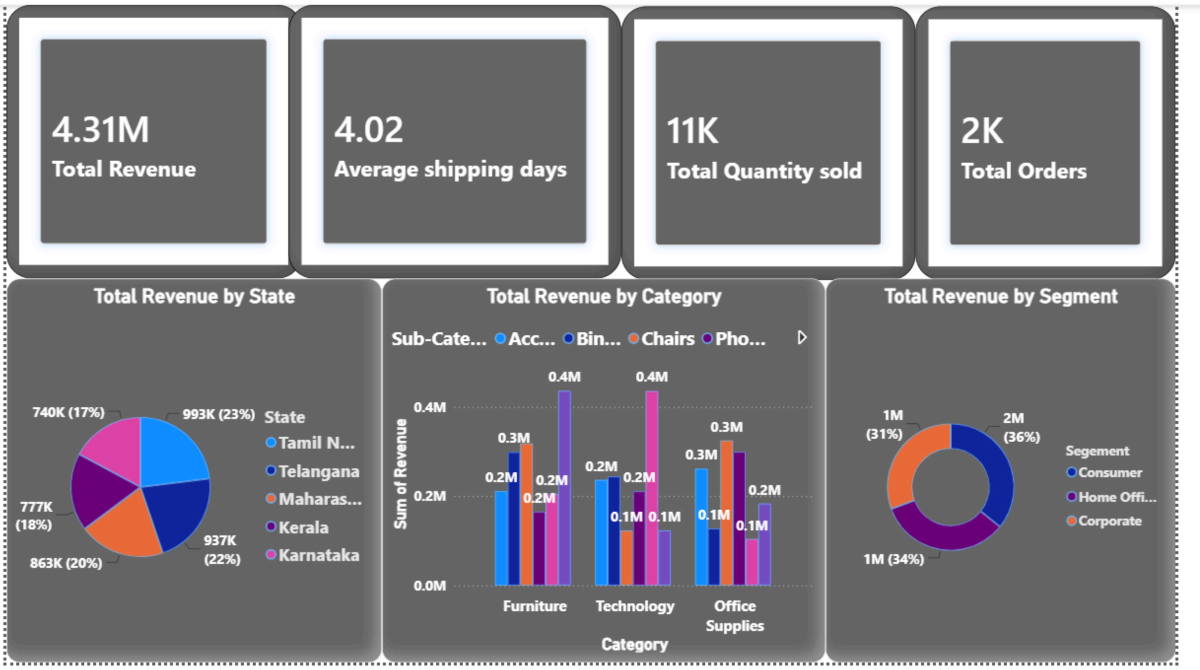
Created a donut chart to show the revenue distribution across different customer segments.

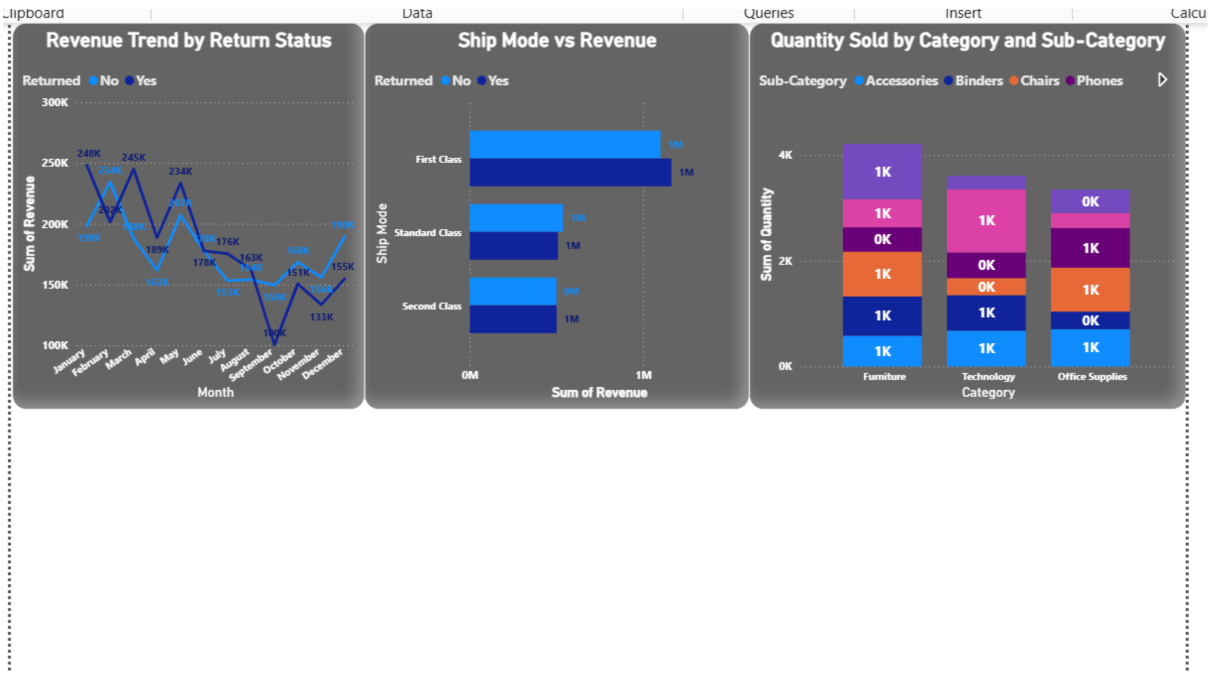
Added a tooltip to display Revenue and Discount details when hovering over the chart

Created a line chart to compare monthly revenue for returned and non-returned orders

Created a clustered bar chart to compare revenue across different ship modes based on return status

Created a stacked column chart to compare quantity sold across categories and sub-categories





9. Insights & Conclusions

Key Findings:

- Identified monthly sales trends.
- Found the best and lowest performing categories.
- Compared sales and profit across different categories.
- Analyzed orders, returns, and shipping modes.

Descriptive Analysis:

Analyzed sales, profit, quantity, and orders.

Diagnostic Analysis:

Identified the reasons for differences in sales and profit.

Predictive Analysis:

Used past sales trends to understand future sales possibilities.

Prescriptive Analysis:

Suggested focusing on high-performing products and reducing returns.

10. Conclusions

The analysis helped to understand sales, profit, orders, and category performance. Power BI made it easy to visualize the data and identify useful business insights.